Master Data Dictionary

Unified Data Dictionary — All Sources

TECHNICAL DOCUMENT

Master's Thesis

Winter Wheat Harvest Window Prediction via Machine Learning

Centre-Val de Loire · France

Indrashil University — 2025

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Version 1.0 — Reference document

Foreword

Intended audience

The **Master Data Dictionary** is the technical reference document of the project. It consolidates in one place the complete and precise description of **all variables** from all data sources used.

It is intended for:

- 👨‍💻 **Developers and data scientists** who will manipulate the data to build the ML models
- 👨‍🔬 **Researchers** who want to understand exactly what each column represents
- 📊 **Analysts** who need to check units, value ranges, and conventions
- 📝 **Thesis writers** who need precise citations
- 🔄 **Future readers** of the project (or of its replication) who want to recreate the data

> **Prerequisite:** this document assumes you have already read the **Agronomic Introduction** (separate pedagogical document). If certain concepts are unfamiliar (NDVI, BBCH, GDD, SAR, etc.), refer to that introduction document.

Document structure

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Notation

In this document, the following conventions are used to characterize columns:

KEY Primary or join identifier **META** Metadata (time, quality, source) **FEATURE** Predictor variable for the model **TARGET** Variable to predict **DERIVED** Computed from other columns

PART I — Overview

1. Project objective and data architecture

1.1 Objective

Objective: predict the **optimal harvest window** of winter wheat at the parcel level, using satellite remote sensing, weather, and soil data. The target is the "50% harvest date" published annually by Céré'Obs (FranceAgriMer).

1.2 Scope

Dimension	Value
Crop	Winter soft wheat (*Triticum aestivum L.*)
Geographic area	Centre-Val de Loire (CVL) region, metropolitan France
Departments covered	6 (Eure-et-Loir 28, Loiret 45, Loir-et-Cher 41, Indre-et-Loire 37, Cher 18, Indre 36)
Study period	2020 → 2024 (5 agricultural campaigns)
Number of parcels	1,500 (300/year × 5 years)
Minimum parcel size	1 hectare

1.3 The 6 data sources

Source	Type	Provider	Role in the project
Céré'Obs	Agricultural phenology	FranceAgriMer + Arvalis	🎯 Target: regional 50%-harvest DOY
RPG	Agricultural cadastre	IGN + Ministry of Agriculture	📍 Parcel identification and geolocation
NASA POWER	Daily weather	NASA Langley Research Center	🌦️ Climate variables (T°, rain, ET0, GDD)
SoilGrids	Soil properties	ISRIC World Soil Information	🪨 Texture, hydraulics, soil fertility
Sentinel-2	Optical satellite	ESA (Copernicus)	🛰️ Vegetation indices NDVI/EVI/NDWI
Sentinel-1 SAR	Radar satellite	ESA (Copernicus)	📡 VV/VH backscatter (all-weather)

1.4 Data architecture diagram

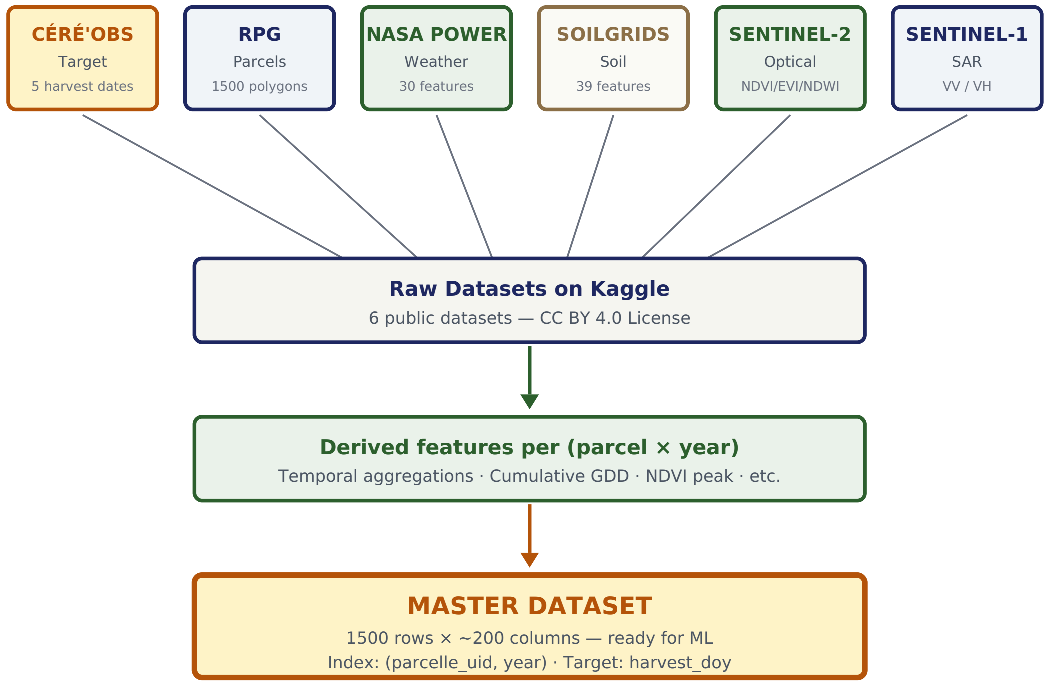

Figure 1.1 — End-to-end architecture: 6 sources → 6 raw public Kaggle datasets → derived features per parcel-year → final master dataset ready for ML.

2. Global conventions

2.1 Column naming conventions

All columns in the project follow these conventions:

Convention	Rule	Example
Case	snake_case (lowercase + underscores)	`peak_ndvi`
English	Names in English (internationally readable)	`harvest_doy`
Source prefixes	Optional based on context	`s2_` , `s1_` , `soil_` , `meteo_`
Statistical suffixes	_median, _mean, _max, _min, _std	`ndvi_median`
Soil depths	_0_5, _5_15, _15_30 (cm)	`clay_0_5`
Weather cumulations	_cumul, _sum	`gdd_cumul`
Day-of-year dates	_doy (Day of Year, 1-365)	`harvest_doy`

2.2 Data types and formats

Pandas type	Project usage	Example
`str` / `object`	Identifiers, codes, labels	parcelle_uid = "2020_4771311"
`int` / `int64`	Year, integer number, code	year = 2020
`float` / `float64`	Continuous measurements (temperatures, indices)	NDVI = 0.847
`datetime64`	Dates or timestamps	scene_date = "2020-05-15"
`bool`	Binary flags	is_valid = True
`geometry`	Polygons / points (GeoPandas)	Shapely WKT point

2.3 Geographic coordinate system

Single reference system: all geographic coordinates in the project are expressed in **WGS84** (EPSG:4326), with longitude (X) and latitude (Y) in decimal degrees. This convention is compatible with Google Earth Engine, web mapping services, and the Sentinel/Copernicus ecosystem.

Data	Original system	Final system
RPG parcels	Lambert 93 (EPSG:2154)	WGS84 (EPSG:4326)
Sentinel-2	UTM 31N (EPSG:32631)	Reused at 20m, aggregated per parcel
Sentinel-1	UTM 31N	Reused at 30m, aggregated per parcel
SoilGrids	WGS84 (EPSG:4326), 250m	Directly used
NASA POWER	WGS84, 0.5° grid	Directly used (6 prefecture points)

2.4 Temporal conventions

Concept	Convention	Example
Harvest year	Year of the harvest July	"season 2020" = Oct 2019 → Jul 2020
Current date	ISO 8601: YYYY-MM-DD	2020-07-03

Day of Year (DOY)	Integer 1-365 (or 366 leap year)	July 1, 2020 → DOY 183
Time zone	UTC for all acquisitions	No local tz in data
Time window	window_start (inclusive)	15j composite starting May 1st

2.5 Unit conventions

Physical quantity	Project standard unit	Notes
Temperature	°C (Celsius)	No Kelvin/Fahrenheit
Precipitation	mm/day	Equivalent to liters/m²
Solar radiation	MJ/m²/day	(Note: NASA POWER provides in MJ/m²/day)
Wind	m/s	No km/h or knots
Relative humidity	% (0-100)	Not fraction (0-1)
Vegetation indices	Unitless, [-1, +1]	NDVI, EVI, NDWI
SAR backscatter	dB (decibels)	Logarithmic scale
Parcel area	hectares (ha)	1 ha = 10,000 m²
Soil depth	cm	Layers 0-5, 5-15, 15-30
Water retention capacity	mm	Equivalent water height
Soil pH	Unitless, 0-14	Logarithmic scale

2.6 Special values

Value	Meaning	Handling convention
`NaN` (float)	Missing data	Standard pandas, impute or filter
`None`	Absence of object (rare in our data)	Converted to NaN in CSV
`-9999`	(Rare) GEE sentinel for nodata	Replace with NaN after reading
`""` (empty string)	Missing text	Converted to NaN

2.7 Locale and CSV separators

Element	Convention
Column separator	`,` (comma, CSV standard)
Decimal separator	`.` (dot, Anglo-Saxon)
Encoding	UTF-8
Headers	First line of the file
Escape characters	`"` around strings containing commas

3. Fusion schema of the 6 sources

The 6 sources have **different granularities**. The following table summarizes how each will be merged to build the master dataset.

3.1 Native granularities

Source	Spatial granularity	Temporal granularity	Cardinality
Céré'Obs	CVL region (1)	Annual (5)	5 valid rows
RPG	Per parcel (1500)	Static	1,500 rows
NASA POWER	Per prefecture (6 points)	Daily (1,766 days)	10,596 rows
SoilGrids	Per parcel (1500)	Static	1,500 rows × 39 features
Sentinel-2 (15j)	Per parcel	15 days (118 windows)	~177,000 rows
Sentinel-1 SAR	Per parcel	Scene-by-scene (~5j)	~3,000,000 rows (5 files)

3.2 Fusion strategy

Principle: we aggregate **all temporal sources** into features per **(parcel × year)**, producing a "long-format" master dataset with 1500 rows (one row per observation) and ~200 columns.

Source	Aggregation method	Features produced
Céré'Obs	Direct join on `year`	1 target column `harvest_doy`
RPG	Direct join on `parcelle_uid`	Metadata: area, department, commune
NASA POWER	Cumulations and statistics over windows (Oct→Jul)	~40 columns (GDD, ET0, cumulative water balance)
SoilGrids	Direct join on `parcelle_uid`	39 columns (13 properties × 3 depths)
Sentinel-2	Derived features: peak_doy, senescence_doy, integral	~30-50 columns
Sentinel-1 SAR	Derived features: vv_min_doy, vh_amplitude, recovery	~30-50 columns

3.3 Join keys

Source	Key(s)	Type
Céré'Obs	`year`	int (2020-2024)
RPG / SoilGrids / S2 / S1	`parcelle_uid`	str (format "YYYY_IDPARCEL")
NASA POWER	`departement` + `date`	str + datetime

Warning — parcelle_uid contains the year: unlike a universal identifier, `parcelle_uid` depends on the year. The physical parcel identified as "2020_4771311" is NOT the same as "2021_4771311". The underlying IGN identifier may be reused from one year to the next by IGN, but we chose to re-sample 300 random parcels each year. Therefore:

- **No temporal continuity** per parcel
- Each (parcelle_uid, year) is an independent observation
- Soil and weather are attached to the physical parcel of the year

3.4 Recommended join method (Pandas)

```
# Pseudocode for final fusion
import pandas as pd

# 1. Load raw sources
cereobs = pd.read_csv('cereobs.csv')           # 5 rows
rpg = pd.read_csv('parcelles_rpg.csv')         # 1500 rows
soil = pd.read_csv('soilgrids.csv')            # 1500 rows
meteo = pd.read_csv('nasa_power.csv')          # 10596 rows
s2 = pd.read_csv('s2_15days.csv')              # 177000 rows
s1 = pd.concat([pd.read_csv(f's1_raw_{y}.csv') for y in range(2020,2025)])

# 2. Aggregate temporal sources into (parcel × year)
s2_features = compute_s2_features(s2)          # 1500 rows × ~30 cols
s1_features = compute_s1_features(s1)          # 1500 rows × ~30 cols
meteo_features = compute_meteo_features(rpg, meteo)  # 1500 rows × ~40 cols

# 3. Final fusion
master = rpg.merge(soil, on='parcelle_uid')
master = master.merge(s2_features, on='parcelle_uid')
master = master.merge(s1_features, on='parcelle_uid')
master = master.merge(meteo_features, on='parcelle_uid')
master = master.merge(cereobs, on='year')      # Join by year

# Result: 1500 rows × ~200 columns
master.to_csv('master_dataset.csv', index=False)
```

PART II — Source-by-source description

5. Source 1 — Céré'Obs (target)

French cereal phenology · Target variable of the project

PROVIDER	TYPE	GRANULARITY	PERIOD
FranceAgriMer + Arvalis	**Observed phenology**	**Regional, annual**	**Since 2015**

5.1 Presentation

Céré'Obs is a French public service that tracks each year the phenological stages of straw cereals (soft wheat, durum wheat, barley, etc.) across the entire territory. Observations are conducted by a network of Arvalis (Institut du végétal) technicians on reference parcels and published weekly and at end of campaign.

For our project, we use the **50%-harvest date** at regional scale (CVL) for each year from 2020 to 2024.

> **Project target variable:** the date when **50% of soft wheat parcels in the CVL region** have reached BBCH stage 92 (mature, harvestable grain). This is the unique target Y that our ML models will attempt to predict from the input features X.

5.2 Source file schema

Original file: `SCR-GRC-CEREOBS_CP_depuis_2015-A26.xlsx`

The Excel file contains the following main columns:

Column (original)	Type	Description
`CAMPAGNE`	str	Year (e.g., "2020"). This is the harvest year.
`CULTURE`	str	Crop type (filter "BTH" = Soft Winter Wheat)
`REGION`	str	Region name (filter "Centre-Val de Loire")
`SEMAINE`	str	ISO week number (e.g., "2020-S27")
`DATE_DEBUT_SEMAINE`	date	First day of the observation week
`SURFACE_RECOLTEE_PCT`	float	% of regional area harvested at that date (0-100)

5.3 Target extraction method

The 50%-harvest date is not explicitly given. We extract it by **linear interpolation** on the weekly series:

1. Filter the file on (CULTURE = "BTH") and (REGION = "Centre-Val de Loire")
2. For each year, identify the two weeks bracketing 50% of harvest
3. Linearly interpolate the date where SURFACE_RECOLTEE_PCT = 50
4. Convert this date to DOY (day of year)

5.4 Result — final values

Year	50%-harvest DOY	Calendar date	Particularity
2020	185	July 3, 2020	Average year
2021	199	July 18, 2021	+14 days (cool spring)
2022	183	July 2, 2022	-2 days (May heatwave)
2023	184	July 3, 2023	Average
2024	195	July 13, 2024	+10 days (wet spring)

5.5 Variables produced for the master dataset

Final column	Type	Unit	Description	Tag
`year`	int	—	Harvest year (2020-2024)	KEY
`harvest_doy`	int	day of year	Regional 50%-harvest DOY	TARGET
`harvest_date`	date	YYYY-MM-DD	Calendar date of 50%-harvest	TARGET

> **Important limitation:** the target is **regional**, not **parcel-level**. All parcels in the same year have the same `harvest_doy` value. Our model therefore predicts the regional average date, exploiting the variability of input features (which are parcel-level). This is a simplification: in an ideal project, we would have true harvest dates per parcel, but such data is not publicly available.

6. Source 2 — RPG (Graphic Parcel Registry)

National agricultural cadastre · Geolocation of 1500 parcels

PROVIDER	TYPE	FORMAT	PERIOD
IGN + Ministry of Agriculture	**Geospatial vector (polygons)**	**Shapefile / GeoPackage**	**Annual, since 2007**

6.1 Presentation

The **Registre Parcellaire Graphique (RPG)** is the French national database that lists all agricultural parcels declared by farmers as part of the CAP (European Common Agricultural Policy). Each parcel is represented by:

- A **geographic polygon** (exact shape of the field)
- The **crop code** (species cultivated that year, e.g., "BTH" for soft wheat)
- A **unique identifier** (per year)
- Metadata (area, INSEE commune code...)

For our project, the RPG serves to **identify and geolocate** the 1500 wheat parcels.

6.2 Sampling strategy

Criterion	Value
Region	Centre-Val de Loire (6 departments)
Crop filter	BTH (Winter Soft Wheat)
Minimum area	1 hectare (to avoid noisy small parcels)
Sampling	Random per year (random_state=42)
Number per year	300 parcels
Total	1,500 parcels (300 × 5 years)

6.3 Final file schema

File: `parcelles_ble_cvl_sample.csv` and `parcelles_ble_cvl_sample.geojson`

Column	Type	Unit	Description	Tag
`parcelle_uid`	str	—	Unique identifier "YYYY_IDPARCEL" (e.g., "2020_4771311")	KEY
`year`	int	—	Agricultural campaign year (2020-2024)	KEY
`id_parcel`	str	—	Original IGN parcel identifier	META
`code_cultu`	str	—	Crop code (always "BTH" in our sample)	META
`surf_parc`	float	hectares	Parcel area (≥1 ha by construction)	FEATURE
`departement`	str	—	INSEE department code (18, 28, 36, 37, 41, 45)	META
`commune`	str	—	INSEE commune code	META
`latitude`	float	WGS84 degrees	Latitude of parcel centroid	META
`longitude`	float	WGS84 degrees	Longitude of parcel centroid	META
`geometry`	Polygon WKT	—	WGS84 polygon (present only in .geojson)	META

6.4 Descriptive statistics

Metric	Value

Total number of parcels	1,500
Mean area	7.9 ha
Median area	5.2 ha
Min / max area	1.0 ha / 89.4 ha
Geographic extent (lat)	46.4°N → 48.9°N
Geographic extent (lon)	0.1°E → 3.1°E

6.5 Validation of annual independence

> **Verification performed:** we verified that **crop rotation** in CVL is sufficiently strong to treat parcels as independent year-on-year. Over 5 years of observation, only **17.1% of parcels** carry wheat in two consecutive years. This validates our independent annual sampling strategy.

7. Source 3 — NASA POWER (daily weather)

Global meteorological reanalyses · Climate variables

PROVIDER	TYPE	GRANULARITY	PERIOD
NASA Langley	**Reanalyses + observations**	**Grid 0.5° × 0.5°**	**Since 1981, daily**

7.1 Presentation

NASA POWER (Prediction Of Worldwide Energy Resources) is a NASA service providing daily global meteorological data, derived from atmospheric reanalyses and satellite measurements. It's free, accessible via API, and covers the entire planet.

For our project, we extract data for **6 points** (one per CVL department, corresponding to the prefecture), over the period **2019-10-01** → **2024-07-31**.

7.2 Extraction points

Department	Prefecture	Latitude	Longitude
28 (Eure-et-Loir)	Chartres	48.45°	1.49°
45 (Loiret)	Orléans	47.90°	1.91°
41 (Loir-et-Cher)	Blois	47.59°	1.33°
37 (Indre-et-Loire)	Tours	47.39°	0.69°
18 (Cher)	Bourges	47.08°	2.40°
36 (Indre)	Châteauroux	46.81°	1.69°

7.3 Raw variables downloaded (8)

NASA code	Description	Unit
`T2M`	Temperature at 2m (daily mean)	°C
`T2M_MIN`	Temperature at 2m (daily minimum)	°C
`T2M_MAX`	Temperature at 2m (daily maximum)	°C
`PRECTOTCORR`	Total corrected precipitation	mm/day
`RH2M`	Relative humidity at 2m	%
`WS2M`	Wind speed at 2m	m/s
`ALLSKY_SFC_SW_DWN`	Solar radiation downward all-sky	MJ/m²/day
`PS`	Surface pressure	kPa

7.4 Derived variables (22)

From the 8 raw variables, we compute **22 agronomic indicators**:

Growing Degree Days (GDD)

Column	Calculation	Unit
`gdd_base0`	(T_max + T_min)/2 − 0 (if > 0)	°C·day
`gdd_base5`	(T_max + T_min)/2 − 5 (if > 0)	°C·day
`gdd_base0_cumul`	Cumulative sum from October 1st	°C·day
`gdd_base5_cumul`	Cumulative sum from October 1st	°C·day

Evapotranspiration and water balance

Column	Calculation	Unit
`et0`	Penman-Monteith FAO-56 formula	mm/day
`et0_cumul`	ET0 cumulated from October 1st	mm
`bilan_hydrique`	PRECTOTCORR − ET0	mm/day
`bilan_hydrique_cumul`	Water balance cumulated from October 1st	mm

Stress indicators

Column	Calculation	Unit
`amplitude_therm`	T_max − T_min	°C
`jours_chauds`	1 if T_max > 30°C else 0	bool
`jours_chauds_cumul`	Cumulative number of hot days	count
`pluie_30j`	Cumulative rainfall over last 30 days	mm
`rayonnement_cumul`	Cumulative radiation from October 1st	MJ/m²

Methodological note — ET0 bug fix: in the first version of the pipeline, we mistakenly applied a `× 3.6` conversion to the ALLSKY_SFC_SW_DWN radiation before computing ET0, assuming it was provided in Wh/m². In reality, NASA POWER provides radiation already in **MJ/m²/day**, so no conversion is necessary. This error, which artificially inflated ET0, has been corrected. The current version uses native values without additional transformation.

7.5 Final file schema

File: `meteo_nasapower_cvl_2020_2024.csv` · 10,596 rows × 30 columns

Column	Type	Description	Tag
`departement`	str	Department code (18, 28, 36, 37, 41, 45)	KEY
`date`	date	Observation date	KEY
(the 8 raw)	float	T2M, T2M_MIN, ..., PS	FEATURE
(the 22 derived)	float	GDD, ET0, water balance, etc.	DERIVED

8. Source 4 — SoilGrids (soil)

Physico-chemical soil properties · 13 variables × 3 depths

PROVIDER	TYPE	RESOLUTION	COVERAGE
ISRIC World Soil Info	**Predictive raster (geostat ML)**	**250 m**	**Worldwide**

8.1 Presentation

SoilGrids is a global database of predictive maps of soil properties, produced by ISRIC (Dutch organization dedicated to soil information). The maps are generated by machine learning from hundreds of thousands of reference soil profiles.

For our project, we extract **13 properties** at **3 depths** (0-5 cm, 5-15 cm, 15-30 cm), at the centroid of each parcel.

8.2 The 13 extracted properties

Property	Code	Unit	Typical range (CVL wheat)
Bulk Density	`bdod`	kg/dm³	1.3 – 1.6
Cation Exchange Capacity	`cec`	cmol(+)/kg	8 – 25
Coarse Fragments (stones)	`cfvo`	cm³/dm³	0 – 150
Clay	`clay`	g/kg (‰)	100 – 350
Nitrogen (total)	`nitrogen`	cg/kg	50 – 200
pH (water)	`phh2o`	pH × 10	55 – 75 (= pH 5.5-7.5)
Sand	`sand`	g/kg	200 – 600
Silt	`silt`	g/kg	250 – 600
Soil Organic Carbon	`soc`	dg/kg	50 – 250
SOC Density	`ocd`	hg/m³	30 – 150
SOC Stock	`ocs`	t/ha	10 – 60
Water Capacity (10 kPa)	`wv0010`	cm³/dm³	250 – 400
Water Capacity (1500 kPa)	`wv1500`	cm³/dm³	100 – 250

Note on SoilGrids units: values are stored as **integers** with a scale factor to save space. For example, pH is multiplied by 10 (so pH 6.5 = value 65). Check scale factors for each variable in the official SoilGrids documentation.

8.3 The 3 depths

Each property is measured at 3 depths representing the wheat **main root zone**:

Depth	Suffix	Interest
0 – 5 cm	`_0_5`	Soil surface — first contact with seeding
5 – 15 cm	`_5_15`	Main rooting zone
15 – 30 cm	`_15_30`	Root system extension zone

8.4 Final schema

File: `soilgrids_features_cvl_1500.csv` · 1500 rows × 41 columns

Column	Type	Description	Tag

`parcelle_uid`	str	Parcel identifier	KEY
`year`	int	Year	KEY
`clay_0_5`	float	Clay (g/kg) at 0-5 cm	FEATURE
`clay_5_15`	float	Clay at 5-15 cm	FEATURE
`clay_15_30`	float	Clay at 15-30 cm	FEATURE
... (same for the other 12 properties × 3 depths = 39 columns total)			

8.5 Extraction methodology (WCS)

> **Methodological optimization:** we initially tested the SoilGrids point-by-point REST API, which would have taken ~5 hours for the 1500 parcels. By switching to **WCS raster extraction** (Web Coverage Service) — downloading the 39 complete rasters then sampling locally — the time dropped to **3.3 minutes**, a **91× speedup**.

8.6 Handling missing values

Out of 1500 parcels, 2 (0.13%) had **NoData** values for certain properties, due to SoilGrids raster artifacts near administrative borders. We applied **nearest-neighbor imputation** (distance 0.22 km = 1 SoilGrids pixel), which is agronomically justified (soil properties vary little at this distance).

9. Source 5 — Sentinel-2 (vegetation indices)

Multispectral optical · NDVI, EVI, NDWI at 3 granularities

PROVIDER	MISSION	RESOLUTION	REVISIT
ESA (Copernicus)	**Sentinel-2A / 2B**	**10-20 m**	**~5 days (S2A+S2B)**

9.1 Presentation

Sentinel-2 is a constellation of two optical satellites (S2A launched in 2015, S2B in 2017) providing high-resolution multispectral imagery of the Earth. The onboard MSI instrument (Multi-Spectral Instrument) captures 13 spectral bands from visible to SWIR.

For our project, we use the **COPERNICUS/S2_SR_HARMONIZED** collection via Google Earth Engine, providing surface reflectance products (Level-2A), atmospherically corrected and harmonized over 2017-today.

9.2 Spectral bands used

Band	Name	Wavelength	Resolution	Usage
B2	Blue	490 nm	10 m	EVI (atmosphere correction)
B4	Red	665 nm	10 m	NDVI, EVI
B8	NIR	842 nm	10 m	NDVI, EVI, NDWI
B11	SWIR	1610 nm	20 m	NDWI

9.3 Computed indices

Index	Formula	Range	Captures
NDVI	(B8 − B4) / (B8 + B4)	[-1, 1]	Greenness, photosynthetic vigor
EVI	2.5 × (B8 − B4) / (B8 + 6×B4 − 7.5×B2 + 1)	[-1, 1]*	Vigor (soil/atmosphere corrected)
NDWI	(B8 − B11) / (B8 + B11)	[-1, 1]	Leaf hydration

> * EVI can produce aberrant values (up to ±67) in case of dark pixels or numerical divergences. **Recommended filtering: |EVI| < 1.5** before any modeling (159 outliers among 131,287 valid values, i.e., 0.12%).

9.4 Extraction pipeline

1. **Collection filtering:** COPERNICUS/S2_SR_HARMONIZED, dates 2019-10-01 → 2024-07-31, intersection with parcels, CLOUDY_PIXEL_PERCENTAGE < 60%
2. **Cloud masking:** Scene Classification Layer (SCL) — exclusion of classes 3 (cloud shadow), 7-9 (cloud probabilities), 10 (cirrus), 11 (snow)
3. **Computation of the 3 indices** per scene
4. **Parcel-level aggregation:** reduceRegions with spatial mean at 20 m
5. **Production of 3 granularities:**
   - 15-day composites (118 windows × 5 statistics) → classical ML
   - 7-day composites (253 windows × 5 statistics) → sequential models
   - Raw per scene, per year (5 files) → maximum flexibility

9.5 Schema of composites (15j and 7j)

Files: `sentinel2_composites_15days_cvl.csv` (177k rows) · `sentinel2_composites_7days_cvl.csv` (~380k rows)

Column	Type	Unit	Description	Tag
`parcelle_uid`	str	—	Parcel identifier	KEY

`year`	int	—	Year (2020-2024)	KEY
`window_start`	date	YYYY-MM-DD	Start of composite window	META
`window_days`	int	days	Window length (7 or 15)	META
`n_images`	int	—	Number of valid scenes in the window	META
`NDVI_median`	float	[-1, 1]	Median NDVI over the window	FEATURE
`NDVI_mean`	float	[-1, 1]	Mean NDVI	FEATURE
`NDVI_max`	float	[-1, 1]	Maximum NDVI (local peak)	FEATURE
`NDVI_min`	float	[-1, 1]	Minimum NDVI	FEATURE
`NDVI_stdDev`	float	≥ 0	NDVI standard deviation (variability)	FEATURE
EVI_* (×5)	float	[-1, 1]	Same 5 stats for EVI	FEATURE
NDWI_* (×5)	float	[-1, 1]	Same 5 stats for NDWI	FEATURE

Total: 19 columns (4 identifiers/meta + 15 features = 5 stats × 3 indices)

9.6 Schema of raw files (5 yearly files)

Files: `sentinel2_raw_2020.csv` ... `sentinel2_raw_2024.csv`

Column	Type	Description	Tag
`parcelle_uid`	str	Parcel identifier	KEY
`year`	int	Year	KEY
`scene_date`	date	S2 scene acquisition date	META
`NDVI`	float	Parcel-mean NDVI for this scene	FEATURE
`EVI`	float	Parcel-mean EVI	FEATURE
`NDWI`	float	Parcel-mean NDWI	FEATURE

Raw files NaN structure: ~92% of rows contain NaN. This is **not** data loss: Google Earth Engine produces one row per (parcel × scene_date), but each S2 scene only covers part of the territory. Parcels outside the scene footprint receive NaN. To get true measurements: `df.dropna(subset=['NDVI'])` → ~80,000 valid measurements per year, median ~42 scenes per parcel.

9.7 Data quality

Indicator	Value
15j composites: NaN rate	25.8% (concentrated in winter)
Critical period (May-July): NaN	5-22% (very good)
EVI outliers	0.12% (filter |EVI| < 1.5)
NDVI ↔ NDWI correlation	r = 0.93 (near-redundancy)
Typical wheat NDVI peak	0.80 - 0.90 (April-May)

10. Source 6 — Sentinel-1 SAR (radar)

Synthetic Aperture Radar · All-weather VV/VH backscatter

PROVIDER	MISSION	RESOLUTION	REVISIT
ESA (Copernicus)	**Sentinel-1A / 1B**	**10 m (IW mode)**	**~6 days (S1A+S1B)**

10.1 Presentation

Sentinel-1 is a constellation of two C-band SAR (Synthetic Aperture Radar) satellites operating at 5.4 GHz. Unlike Sentinel-2 (passive optical), Sentinel-1 emits its own radar waves and measures their echo — it works **in all weather conditions**, through clouds, day and night.

For our project, we use the **COPERNICUS/S1_GRD** (Ground Range Detected) collection via Google Earth Engine, which provides products already calibrated in dB.

10.2 Acquisition configuration

Parameter	Retained value	Rationale
Instrument mode	IW (Interferometric Wide)	Standard for terrestrial observation, 250 km swath
Polarisations	VV + VH	Both transmitted polarisations on land
Orbit	DESCENDING only	Consistent geometry (stable incidence angle)
Resolution	10 m	High-resolution product
Unit	dB (decibels)	GEE native radiometric calibration

10.3 Pre-processing applied

1. **Calibration:** automatic by GEE during ingestion (σ^0 in dB)
2. **Speckle filter:** `focal_median 50 m` per scene, reduces granular noise
3. **Ratio computation:** VV/VH = VV − VH (in dB, subtraction = ratio in linear)
4. **Parcel-level aggregation:** spatial mean at 30 m (oversampling)

10.4 Variables produced

Variable	Type	Typical range (wheat)	Meaning
VV	float (dB)	[-15, -5]	Backscatter in co-polarisation, surfaces and vertical structures
VH	float (dB)	[-22, -12]	Backscatter in cross-polarisation, volume scattering (biomass)
VV_VH_ratio	float (dB)	[4, 12]	Ratio = canopy density indicator

10.5 File schema

Files (5): `sentinel1_sar_raw_2020.csv` ... `sentinel1_sar_raw_2024.csv`

Column	Type	Description	Tag
`parcelle_uid`	str	Parcel identifier	KEY
`year`	int	Year (2020-2024)	KEY
`scene_date`	date	S1 scene acquisition date	META
`VV`	float (dB)	σ^0 in VV polarisation	FEATURE
`VH`	float (dB)	σ^0 in VH polarisation	FEATURE
`VV_VH_ratio`	float (dB)	VV − VH	FEATURE

10.6 Data statistics (validated 2020 season)

Indicator	Value	Verdict
Raw rows	601,500	1500 parcels × 401 scene_dates
Valid rows (non-NaN)	119,491 (19.9%)	Structural (scene footprint)
Valid scenes / parcel	median 100 (max 124)	2.4× denser than S2
VV median (entire season)	-10.7 dB	✅ Expected wheat range
VH median	-18.5 dB	✅ Expected range (VH < VV)
VV/VH ratio median	7.7 dB	✅ Expected range

10.7 Wheat SAR phenological signature

VV evolution during the season directly reflects canopy structure:

Period	Median VV (dB)	Stage	Physical interpretation
Oct-Nov	-9.2 to -8.9	Sowing / tillering	Rough soil + small vegetation → high backscatter
Dec-Feb	-9.0 to -9.6	Slow growth	Partial cover, low biomass
March	-10.7	Stem elongation	Canopy develops, soil signal attenuates
Apr-May	**-13.5**	**Heading/flowering**	**Dense canopy, max attenuation → VV minimum**
June	-12.3	Maturation	Senescence begins, attenuation decreases
July	variable	Harvest	High variability: harvester passage

SAR harvest marker: the **sharp recovery of the VV/VH ratio** (from ~6 to ~8 dB) after its April-May minimum is physically linked to canopy removal by harvesting. This is one of the most promising signals for ML prediction.

10.8 S2 ↔ S1 complementarity

The major strength of Sentinel-1 is its **uniform year-round coverage**:

Month	S2 (scenes/parcel)	S1 (scenes/parcel)	Total S1+S2
December	2.6	8.0	10.6
January	5.2	8.0	13.2
February	2.4	8.0	10.4
April	8.0	8.0	16.0
July	10.8	8.0	18.8

Combined S1+S2 density reaches **~18 scenes/parcel/month on average**, enabling advanced multimodal sequential architectures.

PART III — Master schema and derived features

11. Target schema of the master dataset

11.1 Overview

The **master dataset** is the final file that will be used to train the ML models. It aggregates all 6 sources into a unified tabular structure.

Target structure: 1,500 rows × ~200 columns
1 row = 1 observation (1 parcel × 1 year)
1 column = 1 feature (or the target)
Format: CSV or Parquet

11.2 Column categories

Category	Source	Number of columns	Prefix
Identifiers	—	2	—
Parcel metadata	RPG	~5	—
Target	Céré'Obs	1	harvest_
Soil features	SoilGrids	39	soil_
Derived weather features	NASA POWER	~40	meteo_
Derived Sentinel-2 features	S2	~30	s2_
Derived Sentinel-1 features	S1	~30	s1_
Quality flags	—	~5	flag_
TOTAL	—	**~152**	—

11.3 Master schema overview

Section	Columns (examples)	Tag
Identifiers	parcelle_uid, year	KEY
Metadata	departement, surf_parc, latitude, longitude	META
Target	harvest_doy	TARGET
Soil (static)	soil_clay_0_5, soil_clay_5_15, ..., soil_phh2o_15_30	FEATURE
Weather	meteo_gdd_cumul_apr30, meteo_et0_cumul_jun15, meteo_rainfall_amj	DERIVED
Sentinel-2	s2_peak_ndvi, s2_peak_doy, s2_senescence_doy, s2_ndvi_integral_amj	DERIVED
Sentinel-1	s1_vv_min, s1_vv_min_doy, s1_ratio_recovery_doy	DERIVED
Quality	flag_s2_coverage_ok, flag_s1_coverage_ok, flag_outlier_evi	META

12. Planned derived features

12.1 Principle

Derived features (also called **"engineered features"**) are synthetic indicators computed from raw time series. They concentrate phenological information into a single value per parcel-year, accessible to classical ML models.

> **Reminder:** tree-based models (RF, XGBoost) cannot directly process long time series. For these models, each series must be transformed into a few scalars (derived features). Sequential models (LSTM, Transformer) can work directly on raw series.

12.2 Sentinel-2 derived features (~25)

Feature	Calculation	Meaning
`s2_peak_ndvi`	max(NDVI) over the season	Maximum vigor reached
`s2_peak_doy`	DOY of max NDVI	Date of maximum canopy (heading proxy)
`s2_senescence_doy`	First DOY after peak when NDVI < 0.4	Advanced senescence date
`s2_senescence_slope`	(NDVI_peak − NDVI_sen) / (doy_sen − doy_peak)	Drop speed = maturation speed
`s2_ndvi_integral_amj`	Integral (area under curve) Apr-Jun	Proxy biomass cumulated critical period
`s2_ndvi_april_mean`	NDVI average over April	Early vigor
`s2_ndvi_may_mean`	NDVI average over May	Peak vigor
`s2_ndvi_june_mean`	NDVI average over June	Early maturation vigor
`s2_evi_peak`	max(EVI) over the season	Alternative EVI to NDVI
`s2_ndwi_peak`	max(NDWI) over the season	Peak leaf hydration
`s2_ndwi_drop_doy`	DOY when NDWI drops below 0	Early water stress detection
`s2_n_valid_scenes_amj`	Number of valid scenes Apr-Jun	Useful coverage indicator

12.3 Sentinel-1 derived features (~25)

Feature	Calculation	Meaning
`s1_vv_min`	min(VV) over the season	VV minimum value (max canopy)
`s1_vv_min_doy`	DOY of min VV	Date of max canopy per SAR
`s1_vv_amplitude`	VV_max − VV_min over the season	Variation amplitude
`s1_vv_post_min_slope`	(VV_jul − VV_min) / nb_days	Recovery speed = senescence
`s1_vh_integral_amj`	VH integral over Apr-Jun	Biomass proxy via volume scattering
`s1_ratio_min`	min(VV/VH) over the season	Max canopy density
`s1_ratio_min_doy`	DOY of min ratio	Date of max vegetative canopy
`s1_ratio_recovery_doy`	DOY when ratio recovers +1.5 dB after min	⭐ Direct harvest marker
`s1_ratio_amplitude`	ratio_max − ratio_min over the season	Seasonal amplitude
`s1_vv_mean_amj`	VV average over Apr-Jun	Mean structural vigor

12.4 Weather derived features (~40)

Feature	Calculation	Importance
`meteo_gdd_cumul_apr30`	GDD base 0°C cumulated Oct 1 to Apr 30	Early maturation indicator
`meteo_gdd_cumul_may31`	GDD on May 31	Cumul at heading
`meteo_gdd_cumul_jun30`	GDD on June 30	Cumul just before harvest (direct proxy)
`meteo_rainfall_amj`	Cumulative rainfall April-May-June	Critical water availability
`meteo_et0_cumul_amj`	Cumulative ET0 Apr-Jun	Cumulative evaporative demand
`meteo_bilan_hydrique_amj`	Cumulative (rain − ET0) Apr-Jun	Water balance critical period
`meteo_jours_chauds_mj`	Number of T_max > 30°C days in May-June	Heat-stress risk
`meteo_amplitude_thermique_mean_amj`	Mean thermal amplitude Apr-Jun	Daily thermal stress
`meteo_rayonnement_cumul_amj`	Cumulative radiation Apr-Jun	Energy for photosynthesis

12.5 Important note on derived features

> **Construction of derived features:** these features are **not yet present** in the raw files published on Kaggle. They will be computed at the time of master dataset construction (next project phase). This section therefore serves as a **target specification** rather than a description of an existing file.

13. Quality variables and flags

To ensure traceability and facilitate model debugging, we plan to add **quality flags** to the master dataset:

Flag	Type	Meaning
`flag_s2_coverage_ok`	bool	True if ≥ 50% of 15j composite windows are valid
`flag_s1_coverage_ok`	bool	True if ≥ 80 valid scenes over the season
`flag_soil_imputed`	bool	True if soil values imputed by neighbor (2 parcels)
`flag_outlier_evi`	bool	True if EVI outliers detected during the season
`flag_extreme_year`	bool	True for atypical climatic years (2021, 2022)

PART IV — Appendices

14. Error codes and special values

Value	Possible source	Pre-processing action
`NaN`	All files (missing data)	dropna() or imputation depending on context
`EVI > 1.5`	Sentinel-2 (numerical divergence)	Filter as outlier
`EVI < -1.5`	Sentinel-2	Filter as outlier
`NDVI < -0.2 or > 1.0`	Sentinel-2 (corruption)	Very rare, filter if present
`VV < -30 dB`	Sentinel-1 (water, edge pixel)	Filter if present in bulk
`VV > 0 dB`	Sentinel-1 (specular reflection)	Check (frozen soil, water)
`pH = 0`	SoilGrids (nodata)	Impute by neighbor
`PRECTOTCORR < 0`	NASA POWER (artifact)	Replace with 0

15. Alphabetical index of main columns

Column	Source	Section
`amplitude_therm`	NASA POWER	7.4
`bdod_*`	SoilGrids	8.2
`bilan_hydrique_cumul`	NASA POWER	7.4
`cec_*`	SoilGrids	8.2
`cfvo_*`	SoilGrids	8.2
`clay_*`	SoilGrids	8.2
`code_cultu`	RPG	6.3
`commune`	RPG	6.3
`date`	NASA POWER	7.5
`departement`	RPG / NASA POWER	6.3 / 7.5

`et0` / `et0_cumul`	NASA POWER	7.4
`EVI` / `EVI_*`	Sentinel-2	9.3 - 9.6
`gdd_base0_cumul`	NASA POWER	7.4
`gdd_base5_cumul`	NASA POWER	7.4
`geometry`	RPG (geojson)	6.3
`harvest_date` / `harvest_doy`	Céré'Obs	5.5
`id_parcel`	RPG	6.3
`jours_chauds_cumul`	NASA POWER	7.4
`latitude`	RPG	6.3
`longitude`	RPG	6.3
`n_images`	Sentinel-2	9.5
`NDVI` / `NDVI_*`	Sentinel-2	9.3 - 9.6
`NDWI` / `NDWI_*`	Sentinel-2	9.3 - 9.6
`nitrogen_*`	SoilGrids	8.2
`ocd_*` / `ocs_*`	SoilGrids	8.2
`parcelle_uid`	All (key)	3.3
`phh2o_*`	SoilGrids	8.2
`pluie_30j`	NASA POWER	7.4
`PRECTOTCORR`	NASA POWER (raw)	7.3
`PS`	NASA POWER (raw)	7.3
`rayonnement_cumul`	NASA POWER	7.4
`RH2M`	NASA POWER (raw)	7.3
`sand_*`	SoilGrids	8.2
`scene_date`	S2 raw / S1 raw	9.6 / 10.5
`silt_*`	SoilGrids	8.2
`soc_*`	SoilGrids	8.2
`surf_parc`	RPG	6.3
`T2M` / `T2M_MIN` / `T2M_MAX`	NASA POWER (raw)	7.3
`VH`	Sentinel-1	10.5
`VV`	Sentinel-1	10.5
`VV_VH_ratio`	Sentinel-1	10.5
`window_days`	Sentinel-2 (composites)	9.5
`window_start`	Sentinel-2 (composites)	9.5
`WS2M`	NASA POWER (raw)	7.3
`wv0010_*` / `wv1500_*`	SoilGrids	8.2
`year`	All (key)	3.3

16. Public Kaggle datasets

All raw sources are publicly deposited on Kaggle (rgislikassi account) under CC BY 4.0 license:

Source	Kaggle URL
1. Céré'Obs (target)	`kaggle.com/datasets/rgislikassi/cereb-dataset`
2. RPG parcels	`kaggle.com/datasets/rgislikassi/parcele-ble-cvl-300`
3. NASA POWER weather	`kaggle.com/datasets/rgislikassi/meteo-centre-vale-de-loire-2019-2024-nasapower`
4. SoilGrids	`kaggle.com/datasets/rgislikassi/soil-features-centre-val-de-loire-1500-parcels`
5. Sentinel-2 (indices)	`kaggle.com/datasets/rgislikassi/sentinel-2-indices-cvl-1500-wheat-parcels`
6. Sentinel-1 SAR	`kaggle.com/datasets/rgislikassi/sentinel-1-sar-backscatter-cvl-1500-wheat-parcels`

17. Associated documents

Document	Content	Usage
Agronomic Introduction	Concept popularization (wheat, satellites, ML)	Prior reading for novices
Master Dataset Report	Complete end-to-end methodology	Understand design choices
Individual Data Dictionaries	6 source-specific PDFs	Per-source technical details
Individual Dataset Reports	6 source-specific PDFs	Per-source acquisition methodology
EDA Reports	Sentinel-2 and Sentinel-1 exploratory analysis	Data quality validation

18. Citation and license

Recommended citation:
Kassi R. (2025). *Multi-source Sentinel-1/2 + climate + soil dataset for winter wheat harvest prediction in Centre-Val de Loire, France (2020-2024)*. Master's thesis, Indrashil University. Available on Kaggle: `kaggle.com/rgislikassi`

19. Source credits

Source	Official citation
Sentinel-2	ESA Copernicus Programme — Sentinel-2 Mission
Sentinel-1	ESA Copernicus Programme — Sentinel-1 Mission
Google Earth Engine	Gorelick et al. (2017), Remote Sensing of Environment, 202.
NASA POWER	NASA Langley Research Center (LaRC) POWER Project
SoilGrids	Poggio et al. (2021), SOIL 7:217–240. ISRIC World Soil Information
RPG	IGN + Ministry of Agriculture — Registre Parcellaire Graphique
Céré'Obs	FranceAgriMer + Arvalis Institut du végétal